

## SUMMARY

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Computer Vision Researcher specializing in **Computer Vision and Face Recognition** for **autonomous and in-cabin systems**. Delivered **production-grade ML solutions** at **Mercedes-Benz R&D** and published at **IROS and NeurIPS**. Expertise in translating research into scalable systems, with demonstrated impact including **multi-million-euro cost savings**.

## EXPERIENCE

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- **Mercedes-Benz Research and Development India** Bengaluru, India  
*Computer Vision and Deep Learning Engineer* April 2022 – Present

Member of the Interior-Sensing (IS) Algorithm team, driving development of **computer vision solutions for in-cabin automotive systems**. Contributed to **safety-critical and luxury applications** including **face recognition, depth estimation, and weight prediction**. Owned the pipeline from **proof-of-concept (PoC)** to **production-ready systems**, translating research ideas into deployable solutions under real-world constraints.

- **Core responsibilities**
  - Led end-to-end development of production-grade **in-cabin face recognition systems**, from data strategy and model development to deployment, achieving stringent accuracy and robustness targets.
  - Architected a **dual-level authentication framework** enabling both low-latency personalization and high-assurance security for privacy- and payment-sensitive use cases.
  - Developed **cross-domain face recognition** and IR-to-RGB colorization solutions to bridge RGB/NIR modality gaps and reduce dependency on large-scale RGB data collection.
  - Built deep learning models for occupant analysis, including IR-based **weight estimation**, replacing hardware sensors with software-driven solutions.
  - Collaborated across software, quantization, validation, and platform teams, while driving GCP-based ML infrastructure initiatives and contributing to patentable innovations.
- **Achievements**
  - Achieved **state-of-the-art performance in automotive face recognition**, outperforming industry-leading solutions deployed in airport-grade security systems
  - Introduced an **innovative IR-to-RGB colorization approach**, removing reliance on costly data collection and annotation, delivering **€1.5M+ cost savings** and reducing pipeline turnaround time from **210 days to 2 days (99% reduction)**
- **Technical Knowledge:** CNNs, Transformers, Face Recognition, Contrastive Learning, Knowledge Distillation (KD), Incremental Learning, Open-World Learning, VLM

## PATENTS & INTELLECTUAL PROPERTY

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1. **In-Cabin Face Authentication System** (Filed in Germany and India)  
*Implemented Patent Award; Inventor lead; Enabling secure multi-level authentication in automotive systems*
2. **Intelligent In-Vehicle Hyper-Personalization** (Filed in Germany and India)  
*Co-Inventor; Enhancing user experience by assimilating signals, memory, and preferences*

## PERSONAL PROJECTS

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- **Nutrition Coach — Vision-Language Model for Calorie Estimation** [GitHub](#)  
*PyTorch · Hugging Face Transformers · PEFT/LoRA · TRL · bitsandbytes · Qwen3-VL · Kaggle T4 GPU*
  - Fine-tuned Qwen3-VL-4B with QLoRA (4-bit, LoRA) to estimate calories and macros from food images, outputting structured JSON at 100% schema compliance on held-out data.
  - Ran a controlled experiment comparing two label sources, cutting calorie error to 81.4 kcal MAE — ~34% lower than the base model and reducing carbohydrate error by ~45% on a clean lab-measured holdout set.
  - Demonstrated that fine-tuning on synthetic class-average labels (Food101) degraded accuracy below the un-tuned baseline, while real measured data (Nutrition5k) produced genuine estimation — a data-quality finding with direct modeling implications.

## AWARDS & RECOGNITION

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- **Kaizen (Operational Excellence) Recognition** Mar 2026  
*Selected at department level for innovation leading to significant cost and efficiency gains*
- **World Quality Week Recognition** Nov 2025  
*Presented the benefits, optimizations, and achievements of our solutions*
- **Customer Appreciation Recognition** Sep 2025  
*Recognized by Mercedes-Benz client team; [View Post](#)*
- **Impact Award (Silver)** Feb 2025, 2023  
*Recognized for delivering high-impact ML solutions*
- **Team Excellence Award (Silver)** Oct 2024  
*Awarded for collaborative delivery of production-grade systems*

## PUBLICATIONS

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- **Deepak Singh**, S. N. Rai, K. J. Joseph, R. Saluja, V. N. Balasubramanian, C. Arora, A. Subramanian and C. V. Jawahar, “**New Vehicles on the Road? No Problem - We’ll Learn Them Too**”, 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Kyoto, Japan, 2022. [[Link](#)]
- **Deepak Singh**, S. N. Rai, K. J. Joseph, R. Saluja, V. N. Balasubramanian, C. Arora, A. Subramanian and C. V. Jawahar, “**ORDER: Open World Object Detection on Road Scenes**”, NeurIPS Machine Learning for Autonomous Driving (ML4AD) Workshop, 2021. [[Link](#)]
- **Deepak Singh**, A. Rahane, A. Mondal, A. Subramanian and C. V. Jawahar, “**Evaluation of Detection and Segmentation Tasks on Driving Datasets**”, International Conference on Computer Vision and Image Processing (CVIP), Communications in Computer and Information Science (CCIS), Vol. 1567, Springer, 2021. [[Link](#)]

## EDUCATION

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- **International Institute of Information Technology** Hyderabad, India  
*Master of Science by Research in Computer Science, Specialization in Artificial Intelligence* Aug. 2018 – Dec. 2021

**Master’s Thesis** (Graduate Research Assistant, CVIT): Evaluated and benchmarked SOTA computer vision models for object detection, semantic segmentation, and instance segmentation on Cityscapes, BDD, and IDD datasets to study their behavior in autonomous driving scenarios. Designed ORDER (Open-World Object Detection on Road Scenes), introducing Feature-Mix, focal regression loss, and curriculum learning techniques to improve unknown object recognition and small object detection, achieving superior results on BDD and IDD datasets.